P-values

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Where this fits: Pipeline map step 0 (Foundations). See the Course map.

0 Foundations	1 Frame	2 Get data	3 Understand	4 Clean	5 Features	6 Reduce
7 Split	8 Model	9 Evaluate	10 Tune	11 Deploy	12 Test	13 Monitor

- **Builds on:** Cumulative distribution function (CDF); Hypothesis testing: null and alternative.

1. Overview

Key point: The **p-value** (G-1433) is the probability, if H_0 (the **null hypothesis** (G-1361), the claim that there is no effect) were true, of getting a sample at least as extreme as ours; the smaller it is, the stronger the evidence against H_0.

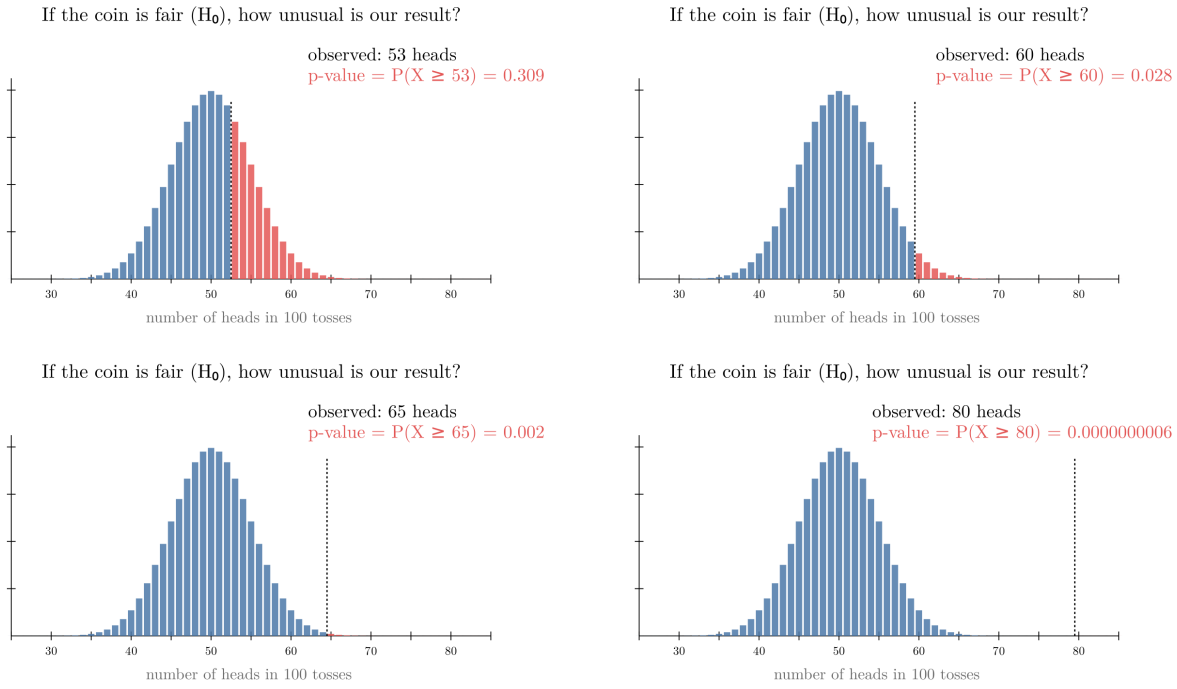


Figure 1: P-value of 100 coin tosses under H_0 (fair coin): the red bars, at or beyond the observed number of heads, add up to the p-value

The rejection region approach only says whether a test statistic (a number computed from the sample, see the z-statistic) crossed a boundary (see the rejection region and the critical value). The **p-value approach** (G-1431) computes one more number, the p-value, which also measures how strong the evidence is. Figure 1 shows the idea: the further the observed result sits in the tail, the smaller the red area, and the smaller the p-value.

This Note covers:

- the definition of the p-value, built on a coin-tossing example;

- how to read a p-value correctly, and the most common misreadings;
- the decision rule $p \leq \alpha$, and a scale of evidence for when no α is given;
- p-values for the one-tailed and two-tailed z-tests of the rejection region.

2. Definition

> **Key point:** The p-value is P(a result as or more extreme than ours | H_0 true), where "more extreme" means "more evidence against H_0".

The **p-value** (G-1433) is the probability of getting a sample as extreme as our own sample, or more extreme, given that the **null hypothesis** (G-1361) is true.

Two words need care:

- **"as or more extreme"**: having as much or more evidence against H_0, in the direction(s) that H_1 points to;
- **"given that H_0 is true"**: the probability is computed in an imaginary world where H_0 holds.

In simple words, the p-value is a measure of the strength of the evidence against the null hypothesis that our sample provides.

2.1 The definition on five tosses

> **Key point:** Four heads in five tosses of a fair coin: the **p-value** (G-1433) is our result plus everything more extreme: 5 outcomes out of 32, plus 1 out of 32, which is about 0.19.

A small case can be counted by hand. We suspect a coin favours heads, toss it 5 times and get 4 heads. H_0 says the coin is fair; H_1 says heads are more likely.

1. **The world of H_0.** Five tosses of a fair coin have 32 equally likely outcomes (2^5). Sorted by the number of heads, they give the bars $1, 5, 10, 10, 5, 1$ out of 32 (Figure 2, first frame).
2. **Our result.** Exactly 4 heads happens in 5 of the 32 outcomes:

$$\frac{5}{32} = 0.156$$

3. **Anything more extreme.** Five heads would be even stronger evidence for a coin that favours heads. It happens in 1 outcome: 1/32.
4. **Add them.**

$$p = \frac{5}{32} + \frac{1}{32} = \frac{6}{32} = 0.19$$

Why add the more extreme results? A single result is always unlikely when there are many possible results: even the most likely count in 100 tosses, 50 heads, has probability 0.08. So

the probability of our exact result says little. The question that matters is how often a fair coin does **at least this well** against H_0.

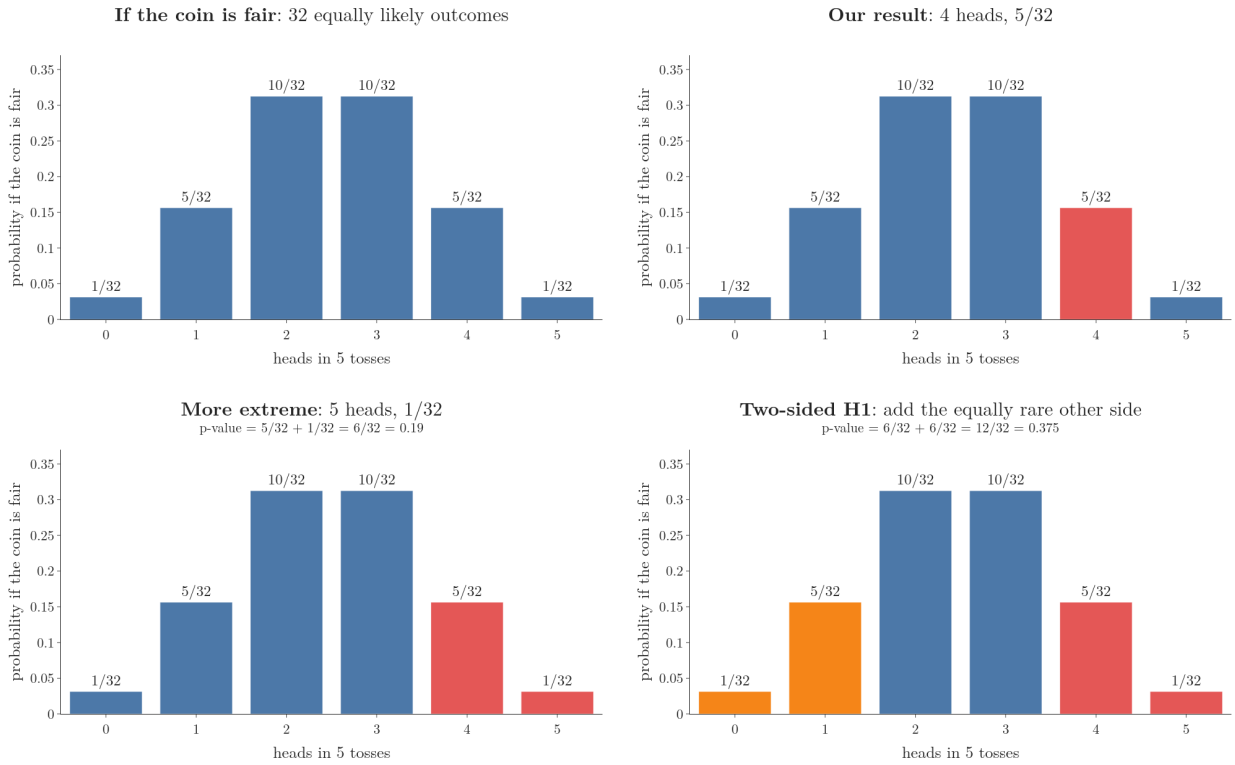


Figure 2: Five tosses of a fair coin. Red: the observed 4 heads, then the more extreme 5 heads; together 6/32. Last frame: a two-sided alternative also counts the equally rare other side (orange), 12/32. Idea after StatQuest, "How to calculate p-values".

The last frame of Figure 2 changes H_1 to "the coin is not fair", with no direction. Then 0 or 1 heads is as much evidence against H_0 as 4 or 5 heads, and both sides are added:

$$\frac{12}{32} = 0.375$$

The direction of H_1 decides which bars count; section 6 does the same for the z-test.

3. A coin tossed 100 times

> **Key point:** For 53 heads out of 100, the **p-value** (G-1433) is $P(X \geq 53) = 0.309$ under a fair coin; for 60 heads it is 0.028, for 80 heads about 6×10^{-10}.

3.1 The experiment

Our experiment: toss a coin 100 times and count the heads. Each toss is a Bernoulli trial (head or tail), repeated $n = 100$ times, so the number of heads X follows a **binomial distribution**

(G-308; see the binomial distribution). If the coin is fair, $X \sim \text{Binomial}(100, 0.5)$, which reads: X follows the binomial distribution with 100 tosses and head probability 0.5.

Figure 1 plots its **PMF** (probability mass function: the probability of each possible count of heads, one bar per count). The PMF looks like a normal curve, but it is discrete. 50 heads is the most likely count. Counts between 40 and 60 come up 96.5% of the time; 65 or more, or 30 or fewer, are rare.

3.2 The hypotheses

We suspect the coin is rigged so that heads come up more often than tails:

$$H_0 : P(\text{head}) = 0.5 \quad \text{(fair coin)}$$

$$H_1 : P(\text{head}) > 0.5 \quad \text{(rigged towards heads)}$$

We toss the coin 100 times and get **53 heads**. One experiment cannot settle the question by looking: 53 is above 50, but is it far enough above?

3.3 The p-value for 53 heads

"As or more extreme than 53" means 53 heads or more: under H_1 the coin favours heads, so more heads is more evidence against H_0.

1. **In words:** add up the probabilities of 53, 54, ..., 100 heads under a fair coin.
2. **Formula:**

$$p = P(X \geq 53 \mid H_0)$$

$$p = \sum_{k=53}^{100} \binom{100}{k} 0.5^{100}$$

   Here $\sum_{k=53}^{100}$ adds the term for $k = 53$, then $k = 54$, up to $k = 100$, and $\binom{100}{k}$ is the number of ways to choose which k of the 100 tosses are heads.

3. **Example:** there are 48 terms, each from the binomial formula. The first six:

k	53	54	55	56	57	58
$P(X = k)$	0.067	0.058	0.048	0.039	0.030	0.022

   The terms shrink towards 0 at 100 heads. The Python box below does the addition:

$$\text{first six} = 0.264$$

$$\text{other } 42 = 0.044$$

$$p = 0.264 + 0.044 = 0.309$$

The probability of exactly 53 heads is not the **p-value** (G-1433). The p-value is the whole tail from 53 upwards: the red bars in Figure 1.

The further the observation is from 50, the smaller the tail:

Heads observed	$P(X = k)$	p-value $P(X \geq k)$
53	0.067	0.309
60	0.011	0.028
65	0.0009	0.0018
80	4×10^{-10}	6×10^{-10}

With 80 heads the red area is invisible: if the coin were fair, 80 heads would almost never happen. Seeing it is overwhelming evidence that the coin is not fair.

> **Python:** Binomial tail probabilities with scipy. `sf(k)` is $P(X > k)$, so $P(X \geq 53)$ is `sf(52)`.

```
from scipy import stats

coin = stats.binom(n=100, p=0.5)
coin.pmf(53)        # 0.067: exactly 53 heads
coin.sf(52)         # 0.309: 53 or more heads, the p-value
coin.sf(59)         # 0.028: 60 or more heads
```

4. Reading a p-value correctly

> **Key point:** $p = 0.309$ means: if the coin were fair, about 31% of repeated 100-toss experiments would give 53 or more heads. The **p-value** (G-1433) is not the probability that H_0 is true.

4.1 The correct reading

Imagine repeating the whole experiment, 100 tosses, many times with a fair coin. A **p-value** (G-1433) of 0.309 says that in about 31% of those experiments we would see 53 heads **or more**. A result like ours is common for a fair coin, so it is weak evidence against H_0. Figure 3 does the repeating: 10,000 experiments of 100 tosses with a fair coin. After 10 experiments the share with 53 or more heads is 30.0%, after 100 it is 32.0%, and after 10,000 it is 31.2%, close to the p-value 30.9%.

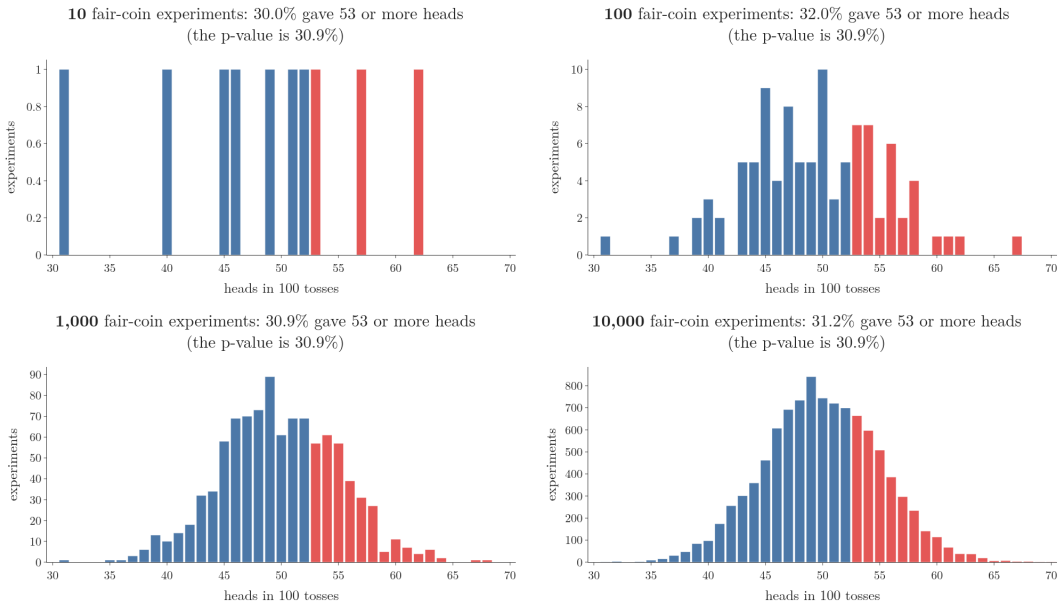


Figure 3: 10 to 10,000 simulated experiments of 100 fair-coin tosses. Red: experiments with 53 or more heads; their share settles near the p-value, 30.9%.

For 60 heads, $p = 0.028$: only about 3 experiments in 100 with a fair coin would give 60 or more heads. Either we were unlucky, or the coin is not fair. For 80 heads, practically no experiment with a fair coin would ever get there.

4.2 Common misreadings

Extra: Each of these statements about $p = 0.03$ is wrong. They are among the misinterpretations listed by Greenland et al. (2016).

- **"About 3 experiments in 100 would give exactly our result."** The **p-value** (G-1433) counts results **as or more extreme**, not exactly ours. For 53 heads, exactly 53 has probability 0.067, while the p-value is 0.309.
- **"There is a 3% chance that H_0 is true."** The p-value is computed **assuming** H_0 is true; it cannot also be the probability of H_0. The probability of H_0 would need **Bayes' theorem** (G-269) and a prior (see the formula of Bayes' theorem).
- **"There is a 97% chance that H_1 is true."** Same mistake, the other way round.
- **"The result happened by chance with probability 3%."** The p-value assumes chance alone (that is what H_0 says); it does not measure the probability of chance.
- **"A small p-value means a large or important effect."** With a huge sample, a tiny difference gives a tiny p-value. The p-value measures evidence, not size: a training program that adds 0.1 cars a day can be "highly significant" and useless. Figure 4 shows it for the right-tailed z-test with $\sigma = 5$: with the true effect fixed at 0.1 cars a day, the p-value drops below 0.05 once the sample passes about 6,765 employees, and keeps falling. Both axes of Figure 4 are logarithmic: each labelled step is ten times the one before (10, 100, 1000 employees; 0.1, 0.01, 0.001 for the

p-value), so a wide range fits in one picture.

- **"A p-value above 0.05 proves H_0."** A p-value above 0.05 means only that the evidence was not strong enough (see the null hypothesis).

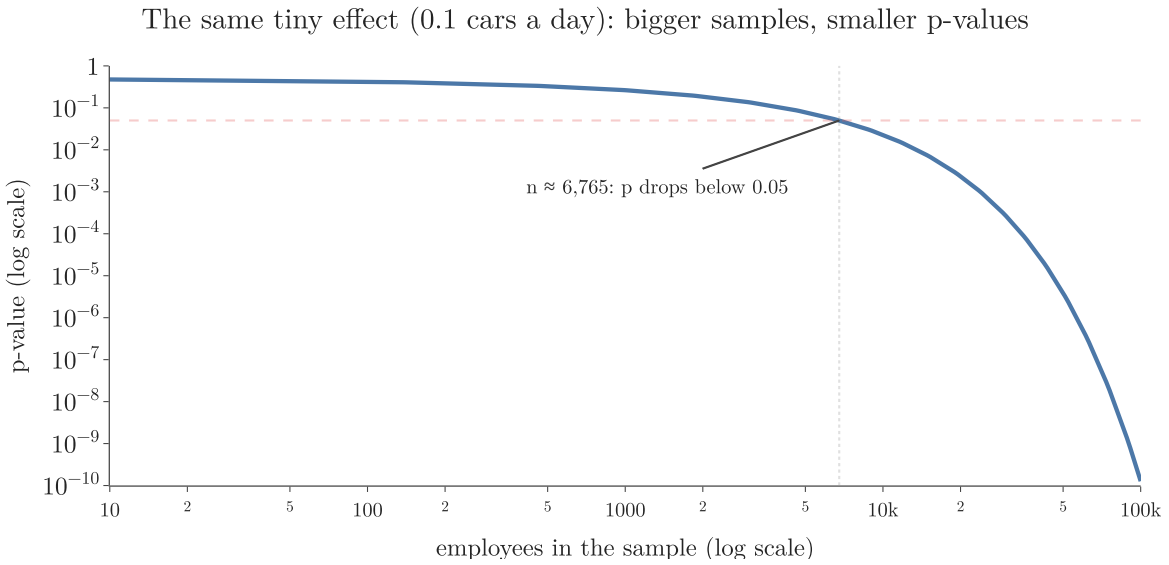


Figure 4: The p-value of the right-tailed z-test at a true effect of 0.1 cars a day ($\sigma = 5$), for samples of 10 to 100,000 employees. The effect never changes; only the sample size does.

5. Deciding with a p-value

Key point: If $p \leq \alpha$, reject H_0; if $p > \alpha$, fail to reject it. Without a fixed α, read the **p-value** (G-1433) on a scale of evidence.

5.1 The decision rule

We still fix the significance level α first, usually 0.05 (see how rare is too rare). Then:

- $p \leq \alpha$: reject H_0;
- $p > \alpha$: fail to reject H_0.

For the coin at $\alpha = 0.05$:

Heads	p-value	Decision	Conclusion
80	0.0000000006	reject H_0	the coin favours heads
60	0.028	reject H_0	the coin favours heads

Heads	p-value	Decision	Conclusion
53	0.309	fail to reject H_0	not enough evidence that the coin is unfair

With 53 heads we cannot conclude that the coin is unfair. We also cannot say we proved it fair.

What does the 0.05 promise? Take a case where H_0 is true by construction: the training-program test of section 6.1 ($H_0 : \mu = 50$, $\sigma = 5$, $n = 30$, right-tailed), run on samples drawn from a population whose mean really is 50. Nothing is there to find, yet the samples differ by chance, and so do their **p-values** (G-1433). Figure 5 collects them. With H_0 true, the p-values spread evenly between 0 and 1, so about 5 percent of them fall at or below 0.05: 5.3 percent of the 10,000 tests here. Each of those is a **false positive** (G-748): the rule rejects a true H_0. So $\alpha = 0.05$ means that when there is no effect, the rule raises a false alarm in about 1 test out of 20. A stricter α of 0.01 would cut the false alarms to 1 in 100.

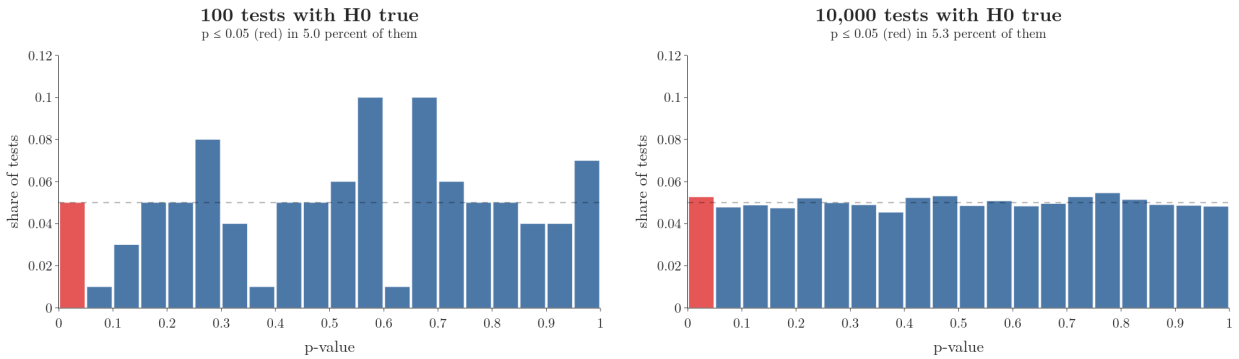


Figure 5: P-values of 100, 1,000 and 10,000 z-tests on samples drawn with H_0 true. The bars level out at 5 percent each (dashed line); the red bar, $p \leq 0.05$, holds the false positives. Idea after StatQuest, "p-values: What they are and how to interpret them".

> **Extra:** Why the rule $p \leq \alpha$ gives the same decision as the rejection region. The rejection region is the tail whose area is α; the p-value is the tail area beyond our statistic. Our statistic lies inside the rejection region exactly when it is further out than the critical value, that is, exactly when the tail beyond it is smaller than the tail beyond the critical value: $p \leq \alpha$. The two approaches always agree on the decision; the p-value adds how far inside or outside we landed.

5.2 A scale of evidence

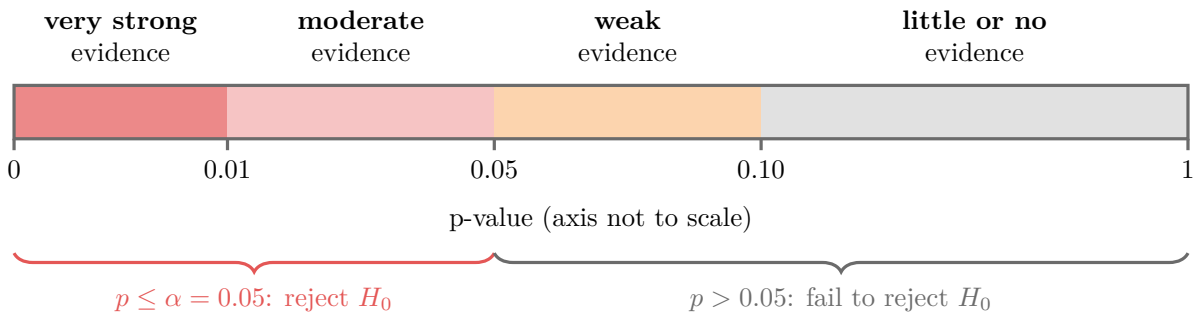


Figure 6: A common rule of thumb for reading p-values

Sometimes no α is given, for example when we explore an independent dataset in a machine learning project. Then a rule of thumb helps (Figure 6):

p-value	Evidence against H_0	What to do
below 0.01	very strong	reject H_0
0.01 to 0.05	moderate	reject H_0
0.05 to 0.10	weak	investigate further: more data, domain knowledge, talk to stakeholders
above 0.10	little or none	fail to reject H_0

In practice most people still use $\alpha = 0.05$, a balance between Type I and Type II errors (see Type I and Type II errors).

6. P-values for the z-test

> **Key point:** Compute z as before; the **p-value** (G-1433) is the tail area beyond z in the direction of H_1, doubled for a two-tailed test. No critical value is needed.

The **p-value approach** (G-1431) works for continuous distributions in the same way: the sum of bars becomes an area under the curve. We redo the two z-tests of the training program and the chips packets, with the same data.

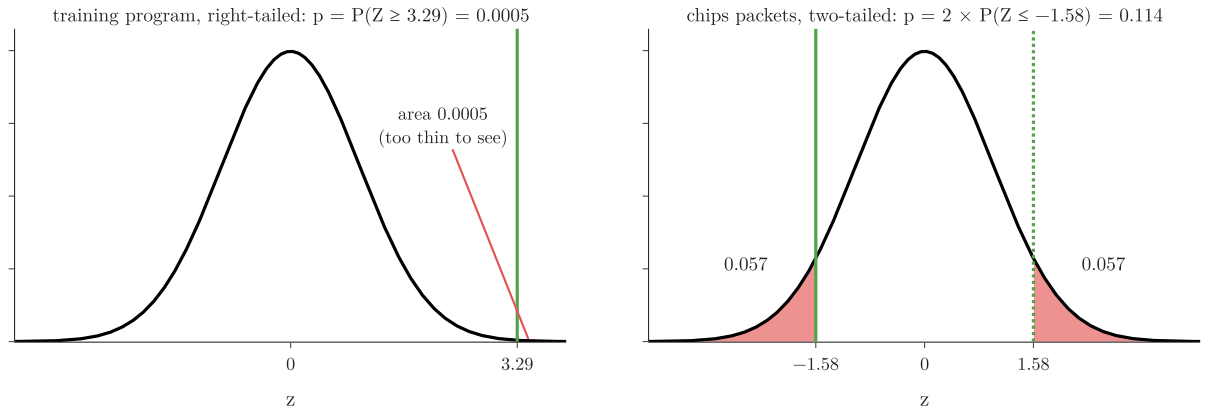


Figure 7: P-values (red areas) of the two z-tests

6.1 One-tailed: the training program

$H_0 : \mu = 50$ (the population mean is 50), $H_1 : \mu > 50$, $\sigma = 5$ (the population standard deviation), $n = 30$ (the sample size), $\bar{x} = 53$ (the sample mean), $\alpha = 0.05$. The z statistic is $z = 3.29$, as before.

1. **In words:** H_1 points right, so "more extreme" means a larger z; the **p-value** (G-1433) is the area to the right of 3.29.
2. **Formula:** the z-table gives the area to the left of z under the standard normal curve, written $\Phi(z)$, so
$$p = P(Z \geq z) = 1 - \Phi(z)$$
3. **Example:**
$$p = 1 - \Phi(3.29) = 1 - 0.9995 = 0.0005$$

$0.0005 \leq 0.05$: we reject H_0. The training program raised productivity, and the evidence is very strong (Figure 7, left).

Compare a smaller z of 1.7:

$$p = 1 - \Phi(1.7)$$

$$p = 1 - 0.9554 = 0.045$$

This p-value is also below 0.05, so both samples lead to "reject". But 0.045 is close to the boundary, while 0.0005 leaves no doubt. This difference is exactly what the rejection region approach could not show.

6.2 Two-tailed: the chips packets

$H_0 : \mu = 50$, $H_1 : \mu \neq 50$, $\sigma = 4$, $n = 40$, $\bar{x} = 49$, $\alpha = 0.05$. The z statistic is $z = -1.58$.

1. **In words:** H_1 has no direction, so results far out on **either** side count as more extreme. We take the area beyond -1.58 on the left and its mirror image beyond $+1.58$ on the right.
2. **Formula:** by symmetry the two areas are equal:

$$p = P(Z \leq -|z|) + P(Z \geq |z|)$$

$$p = 2\,\Phi(-|z|)$$

3. **Example:** the z-table gives $\Phi(-1.58) = 0.057$:

$$p = 2 \times 0.057 = 0.114$$

$0.114 > 0.05$: we fail to reject H_0 (Figure 7, right). The 40 packets do not show that the mean weight differs from 50 g.

6.3 Beyond the z-table

A printed z-table stops around $z = 4$, where the area is practically 1. For larger values, software gives the exact tail. For $z = 15$:

$$P(Z \geq 15) = 4 \times 10^{-51}$$

Python: P-values (G-1433) for z statistics. `sf` (survival function) is 1 − CDF: the right-tail area.

```
from scipy import stats

stats.norm.sf(3.29)              # 0.0005: right-tailed
stats.norm.cdf(-1.645)           # 0.05:   left-tailed
2 * stats.norm.cdf(-abs(-1.58)) # 0.114:  two-tailed
stats.norm.sf(15)                # 3.7e-51
```

7. Summary

	Rejection region approach	P-value approach
Computes	test statistic and critical value	test statistic and p-value
Decision	statistic in the rejection region	$p \leq \alpha$
Strength of evidence	not shown	the smaller p, the stronger
Training program ($z = 3.29$)	$3.29 > 1.645$: reject	$p = 0.0005$: reject
Chips packets ($z = -1.58$)	$\lvert -1.58 \rvert < 1.96$: fail to reject	$p = 0.114$: fail to reject

- The table's "Strength of evidence" row is the reason to prefer **p-values** (G-1433): both approaches reach the same decision, but only p shows that 0.0005 leaves no doubt while 0.045 is close to the boundary.
- The p-value is the probability, under H_0, of a result as or more extreme than ours, so the smaller it is, the harder our sample is to explain by chance alone.
- 53 heads in 100 tosses: $p = 0.309$, so 31% of fair-coin experiments would do at least as well, which makes it weak evidence that the coin is unfair.
- The p-value is not the probability that H_0 is true, because it is computed assuming H_0 is true; and it does not measure the size of an effect, because with a huge sample a tiny, useless effect gives a tiny p-value.
- Reject H_0 when $p \leq \alpha$; the decision always matches the rejection region approach, because z lies in the rejection region exactly when the tail beyond it is smaller than α.
- One-tailed: the tail beyond z on the side of H_1. Two-tailed: twice the tail beyond $|z|$, because with no direction in H_1 results far out on either side count as more extreme.
- This answers the opening question: the p-value is the chance, if H_0 were true, of a sample at least as extreme as ours; the smaller it is, the stronger the evidence against H_0.

8. Sources

Built from

- CampusX, "Session 46 - Hypothesis Testing Part 2 | p-values | t-tests | DSMP 2023", YouTube, https://www.youtube.com/watch?v=xHTMjxx14sU
- StatQuest with Josh Starmer, "How to calculate p-values", YouTube, https://www.youtube.com/watch
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- StatQuest with Josh Starmer, "p-values: What they are and how to interpret them", YouTube, https://www.youtube.com/watch?v=vemZtEM63GY

Other references

- Greenland, S., Senn, S. J., Rothman, K. J., Carlin, J. B., Poole, C., Goodman, S. N. and Altman, D. G. (2016). "Statistical tests, P values, confidence intervals, and power: a guide to misinterpretations." *European Journal of Epidemiology* 31(4), 337–350.

9. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
P-value (G-1433)	The probability, assuming H_0 is true, of getting a sample as or more extreme than ours.
More extreme (G-1263)	Said of a possible test result with at least as much evidence against H_0 as the observed one, in the direction(s) H_1 points to; the p-value is the probability of a result this extreme or more extreme when H_0 is true.
One-tailed p-value (G-1384)	The p-value counted on one side only: the tail area beyond the test statistic on the side that H_1 points to.
P-value decision rule (G-1432)	Reject H_0 if $p \le \alpha$, otherwise fail to reject it.
Two-tailed p-value (G-2027)	The p-value of a two-tailed test: the chance, if H_0 is true, of a test statistic at least as far from 0 as ours in either direction; for a z-test it is $2\,\Phi(-\lvert z\rvert)$.
Null hypothesis (H_0) (G-1361)	The statement of no effect, no difference or no relationship; assumed true until the data gives strong evidence against it.
P-value approach (G-1431)	Carrying out a test by computing a p-value, which also measures the strength of the evidence.
Binomial distribution (G-308)	The distribution of the number of successes in n independent trials with the same success probability.
Bayes' theorem (G-269)	The rule that reverses a conditional probability: from $P(B \mid A)$ it gives $P(A \mid B)$, so a belief about A can be updated after seeing B; $P(A \mid B) = P(B \mid A)P(A)/P(B)$.
False positive (FP) (G-748)	Predicted positive, but actually negative; a Type I error.